Here are some errata for the OpenStax Chemistry 2e book chapters 3 and 4

https://openstax.org/details/books/chemistry-2e

Small Discrepancies - Atomic Weights, Rounding, and Unknown

For many exercise answers in chapters 3 and 4, I get answers very close to the book answers, but I am unable to exactly match the book answers. I believe the issue may lie in the book answers, not my own, for the following reasons:

- The fact that my answers are close argues that my fundamental approach is sound.
- I have reviewed the rounding and significant digit rules found in Ch 1.5 section *Significant Figures in Calculations*. They did not enable me to exactly match the book answers. In a couple cases, I was able to match the book answers by ignoring those rules (see table below).
- I followed the advice in Ch 3.3 right after example 3.17 which says “refrain from rounding any intermediate calculation results” and “rounding errors may also be avoided by performing computations in a single step”. That does not let me exactly match all book answers. In one case, I was able to match the book answer by ignoring that advice (see table below).
- For some exercises I can only match the book answers if I use atomic weights from sources other than Appendix A in the book. One source that I tried was an International Union of Pure and Applied Chemistry’s periodic table PDF at https://iupac.org/what-we-do/periodic-table-of-elements/. Another source I tried was the chempy python package. (I did not do the exercises with chempy. I only used it as a source for atomic weights). The Appendix A atomic weights are uniformly rounded to 4 significant digits. The atomic weights from the other sources are not. Neither source I cited may be the actual source of atomic weights used, but using those alternate atomic weight sources helps explain around half of the discrepancies. I assumed we were supposed to use the atomic weights from Appendix A.
  Note: Requiring students to use the atomic weights from Appendix A might help reveal students using AI chatbots to help with their homework.
- One example, 3.6, directly includes an atomic weight that does not match Appendix A.
- I redid some problems with a Python Jupyter Notebook, defining constants for the atomic weights. This should have eliminated typos, or exposed them by making every answer wrong. The discrepancies with the book answers persisted.

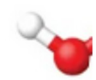
Below I detail discrepancies in examples. Then I detail discrepancies in a couple of exercises. Finally I list all discrepancies in odd exercises in a table. Only odd exercises are included because I do not have instructor access.

Descriptions of Discrepancies in Examples

Ch 3.1, Example 3.6, second unlabeled figure, and its alt text.

In both the figure and alt text, the molar mass of Nitrogen is given in row 5, column 4 as 14.007. This carries through to the subtotal in column 6 of the same row. The atomic weight 14.007 matches the atomic weight of Nitrogen given in the IUPAC periodic table PDF. However, it does not match the atomic weight of Nitrogen given in the periodic table in Appendix A, Figure A1 of the book, which is 14.01.

Element	Quantity (mol element/ mol compound)		Molar mass (g/mol element)		Subtotal (g/mol compound)
C	2	×	12.01	=	24.02
H	5	×	1.008	=	5.040
O	2	×	16.00	=	32.00
N	1	×	14.007	=	14.007
		Molecular mass (g/mol compound)			75.07



Ch 3.4 Example 3.24.
I don’t understand why the answer to this example, 195 g, is rounded to 3 significant digits instead of being rounded to 2 significant digits as 200 g. My reasoning is that multiplication is the only operation, and one of the multiplands, 70%, is only given to two significant digits. Compare with the Check Your Learning problem at the bottom of the example, where the concentration of ethanol is given as 12% and the answer, 1.5 mol, is rounded to 2 significant digits.

Example 4.13, Solution work.
The molar weight of $CuSO_4$ in an intermediate step appears off.

$$1.274\ \cancel{\text{g CuSO}_4} \times \frac{1\ \cancel{\text{mol CuSO}_4}}{159.62\ \cancel{\text{g CuSO}_4}} \times \frac{1\ \text{mo}}{1\ \text{mol C}}$$

Calculating the molar weight of $CuSO_4$ based on the periodic table in Appendix A gives Cu + S + O x 4, which is 63.55 + 32.06 + (16.00 x 4), which is 159.61 g, not 159.62 g.

Example 4.15 Solution work.
I believe the molar mass of $BaSO_4$ is wrong. Part of the equation below the flowchart says it is 233.43 g.

$$0.6168\ \cancel{\mathrm{g\ BaSO_4}} \times \frac{1\ \cancel{\mathrm{mol\ BaSO_4}}}{233.43\ \cancel{\mathrm{g\ BaSO_4}}} \times \frac{1}{1}$$

However, calculating the molar mass directly from the periodic table in Appendix A gives 233.36 g. This also slightly affects the rest of the mass. The book gives the mass of $MgSO_4$ at the end of that equation as 0.3181 g. I believe it should be 0.3182 g. The book gives the final mass concentration to be 69.91 %. I believe it should be 69.92 %.

Descriptions of Discrepancies in Two Exercises

Exercise 3.3 (a) solution.
The problem is to find the molar mass of P_4. Appendix A gives the atomic weight of P as 30.97, which multiplied by 4 gives a solution of 123.88 amu. However, the book gives a solution of 123.896 amu. If you use the atomic weight of P given by the IUPAC periodic table PDF, 30.974, you do indeed get an answer of 123.896 amu.

Exercise 3.3 (b) solution.
The problem is to find the molar mass of H_2O. Appendix A gives the atomic weights as 1.008 for H and 16.00 for O. (2 x 1.008) + 16.00 = 18.016 amu unrounded, which when rounded to the two decimal places present in the O atomic weight would make the solution 18.02 amu. However, the book gives a solution of 18.015 amu. If you use the IUPAC PDF atomic weights of 1.0080 for H and 15.999 for O, you do get the book answer of 18.015 amu.

Table of All Discrepancies in Odd Exercises

Notes:
- I was unable to check even numbered exercises as I am not an instructor.
- The IUPAC atomic weights referred to are those found in their periodic table PDF downloaded from https://iupac.org/what-we-do/periodic-table-of-elements/. I believe this matches the abridged, not full, weights from the tables in their detailed academic papers.
- The chempy python package atomic weights are from release 0.10.1, Sept 22, 2025. The weights may indirectly come from IUPAC unabridged atomic weights.

Exercise	My answer	Book answer	How to get book answer
3.3 a	123.88 amu	123.896 amu	Use IUPAC atomic weights
3.3 b	18.016 amu	18.015 amu	Use IUPAC or chempy weights
3.3 c	164.10 amu	164.086 amu	Use IUPAC or chempy weights
3.3 d	60.05 amu	60.052 amu	Use IUPAC or chempy weights
3.3 e	342.30 amu	342.297 amu	Use IUPAC or chempy weights

Exercise	My answer	Book answer	How to get book answer
3.5 a	56.10 amu	56.107 amu	Unknown
3.5 b	54.09 amu	54.091 amu	Unknown
3.5 c	200.00 amu	199.9976 amu	Unknown
3.5 d	97.99 amu	97.9950 amu	Unknown
3.13 b	72.15 g mol^{-1}	72.150 g mol^{-1}	Unknown
3.13 c	378.10 g mol^{-1}	378.103 g mol^{-1}	Unknown
3.13 d	58.08 g mol^{-1}	58.080 g mol^{-1}	Use IUPAC or chempy weights
3.13 e	180.16 g mol^{-1}	180.158 g mol^{-1}	Unknown
3.15 a	197.38 g mol^{-1}	197.382 g mol^{-1}	Unknown
3.15 b	257.15 g mol^{-1}	257.163 g mol^{-1}	Unknown
3.15 c	194.20 g mol^{-1}	194.193 g mol^{-1}	Unknown
3.15 d	60.06 g mol^{-1}	60.056 g mol^{-1}	Use IUPAC or chempy weights
3.15 e	306.45 g mol^{-1}	306.464 g mol^{-1}	Unknown
3.19 a	99.40 g	99.41 g	Unknown
3.23 Si mass	9.506 g	9.504 g	Use IUPAC or chempy weights
3.33 a	% N = 82.25%, % H = 17.75%;	% N = 82.24%, % H = 17.76%;	Use IUPAC or chempy weights
4.43 c mass	6.80×10^2 g $NaNO_3$	6.8×10^2 g $NaNO_3$	Ignore that the 128 g input has 3 significant figures
4.43 d moles	1670 mol CO_2	1665 mol CO_2	Ignore that the 20.0 kg input has

Exercise	My answer	Book answer	How to get book answer
			3 significant figures
4.43 f moles	0.4581 mol $C_2H_4Br_2$	0.4580 mol $C_2H_4Br_2$	Use IUPAC or chempy weights
4.45 b mass	0.08642 g O_2	0.08644 g O_2	Use IUPAC or chempy weights
4.45 e	16.31 mol BaO_2, 2761 g BaO_2	16.31 mol BaO_2, 2762 g BaO_2	Use chempy atomic weights and use the rounded first answer as input to the second calculation
4.45 f	0.207 mol C_2H_4, 5.82 g C_2H_4	0.207 mol C_2H_4, 5.81 g C_2H_4	Use the rounded first answer as input to the second calculation
4.47 part 2	1.2 mol $GaCl_3$, 2.2×10^2 g $GaCl_3$	1.25 mol $GaCl_3$, 2.2×10^2 g $GaCl_3$	Ignore that the 2.6 L input has 2 significant figures
4.49 a	5.336×10^{22} molecules	5.337×10^{22} molecules	Use chempy weights, or use IUPAC weights and full $6.02214076 \times 10^{23}$ value for N_A
4.49 b	10.40 g $Zn(CN)_2$	10.41 g $Zn(CN)_2$	Use chempy weights, or use IUPAC weights and rounded 0.08862 mol intermediate value in second calculation

Other Issues

Ch 1

I discovered one issue back in Chapter 1.

Ch 1.1

Figure 1.5 alt text.
The end of the first sentence seems grammatically incorrect. It reads “Figure A shows a photo of an iceberg floating in a **sea has three arrows**.” One way to fix it might be to move the “three arrows” to the next sentence, so the first two sentences would read “Figure A shows a photo of an iceberg floating in a **sea. Three arrows point to** figure B, which contains three diagrams showing how the water molecules are organized in the air, ice, and sea.”

Ch 3

Ch 3.1

Regarding the first five figures in Ch 3.1, Figures 3.2, 3.3, unlabeled Example 3.1 figure, Figure 3.4, and unlabeled Example 3.2 figure, the figures containing both a table and an image.
The alt texts for these figures are impressively detailed. However, wouldn't it be even more accessible if each figure was broken up into a real HTML/PDF table, and then a figure. That way table accessibility tools could help users navigate the table portions. It would be ideal if the table and the figure could still be side by side in the visual layout. Just a thought. Perhaps this was already considered and rejected as too complex to implement.

Figure 3.2 alt text.
The end of sentence 4 reads '...and a **blank**, merged cell that runs the width of the first five columns.' Sentence 11 refers to the same merged cell, saying 'The merged cell under the first five columns reads "Molecular mass."' Visually, the latter sentence is correct. That is, the merged cell is not blank. Therefore, should the word "blank," be removed from sentence 4?

Figure 3.2 alt text, Figure 3.3 alt text, and Example 3.1 unlabeled figure alt text.
Would the purpose of the "Molecular mass" label be clearer if sentence 11 about the merged cell was expanded to read 'The merged cell under the first five columns reads "Molecular mass" **on its right side, next to the number in the sixth column.**'
The Figure 3.4 alt text and Example 3.2 unlabeled figure alt text have the same issue, but the merged cell contents say "Formula mass" instead of "Molecular mass".

Figure 3.2, and 3.3 alt text (contd.)
Sentence 12 reads 'To the **left** of the table is a diagram of a molecule.' The word "left" should be "right", instead. The Figure 3.4 alt text has the same issue, but the sentence ends in the words "chemical structure" instead of "molecule".

Figure 3.3 alt text (contd.)
Sentence 6 reads 'The second column contains the numbers "9," "8," and "4" as well as the merged, cell.' The comma before the final word "cell" looks unnecessary.

Figure 3.3 alt text (contd.)
A sentence reads "Attached to each of **the** four black spheres is one smaller white sphere." It is referring to six black spheres in a ring from the previous sentence. Therefore, would it be better to word it "Attached to each of ~~**the**~~ four black spheres is one smaller white sphere."

Figure 3.3 alt text (contd.)
A sentence reads "Attached to the last black sphere of that row are **two** more white spheres." In fact, the last black sphere has three white spheres attached, making it "Attached to the last black sphere of that row are **three** more white spheres."

Figure 3.3 alt text (contd.)
The final sentence reads “A black sphere, attached to two red spheres and a white sphere is attached to the black sphere on the top right of the six-sided ring.” While this is correct, its construction is not parallel to the sentences describing the attachments to the other ring spheres. Would it be clearer as “Attached to the final black sphere in the six-sided ring is another black sphere. Attached to that black sphere are two red spheres, one of which has a smaller, white sphere attached to it.”

Example 3.1 unlabeled figure alt text (issue 1)
Sentence 1 reads “Click to enlarge image of A table is shown that is made up of six columns and five rows.” That seems ungrammatical, and the “A” being capitalized is odd because there is no table “A”. Would it be clearer starting with the same two sentences as the preceding, similar images, that is, “A table and diagram are shown. The table is made up of six columns and five rows.” The similarity with the wording of the preceding images would also give the reader an early hint that this figure is of the same form as those preceding ones. If it is important to keep the “Click to enlarge image” phrase, perhaps that could just be a separate leading sentence “Click to enlarge image.” The word “A:” being capitalized before the word “table” in the existing first sentence makes it seem like that was the original intent anyway.

Example 3.1 unlabeled figure alt text (issue 2)
A sentence reads “The black sphere on the left side of the six-sided ring is **connect** to another black sphere.” The word “connect” should be “**connected**”.

Example 3.1 unlabeled figure alt text (issue 3)
A later sentence dealing with the **left** side of the structure reads “Each of these last two black spheres is connected to **two** smaller, white spheres.” It should read “Each of these last two black spheres is connected to **three** smaller, white spheres.”

Example 3.1 unlabeled figure alt text (issue 4)
An even later sentence dealing with the **right** side of the structure reads “The black sphere that is connected to it and is situated to the top right is connected to **two** smaller, white spheres.” It should read “The black sphere that is connected to it and is situated to the top right is connected to **three** smaller, white spheres.”

Figure 3.4 or Figure 3.4 alt text.
Figure 3.4 does not have “(amu)” below the word “Subtotal” in row 1, column 6 of the table portion. This is unlike the preceding, similar figures. It is also unlike figure 3.4’s own alt text, which gives the text of the sixth header cell as “Subtotal (a m u),” Suggest adding “(amu)” below Subtotal to the figure or removing “(a m u)” from the alt text for consistency.

Example 3.2 unlabeled figure alt text.
The 2nd to last sentence reads “It shows yellow and grey **sphere** connected to red spheres in a complex pattern.” The first “sphere” should be plural, as “It shows yellow and grey **spheres** connected to red spheres in a complex pattern.

Figure 3.6 alt text.
The first sentence reads “This photo shows two vials filled with **a** colorless **liquid**.” However, the figure caption gives context that they are not the same colorless liquid (1-octanol vs methanol). Would the sentence be better as “This photo shows two vials filled with colorless liquid**s**.”

Example 3.6, second unlabeled figure, alt text.
Sentence 2 describing the header row of the table portion ends with “Subtotal (a m u).” In this case it should be “Subtotal (g / mol compound)”.

Example 3.6, second unlabeled figure, **and** its alt text.
The alt text sentence describing the merged cell contents ends “Molar mass (g / mol compound). This does not exactly match the cell contents in the figure (Molar vs Molecular). It is also missing a closing quotation mark. It should read “Molecular mass (g / mol compound)” to match the figure. I do not know which is correct but they are inconsistent.

Ch 3.4

Example 3.24.
I don’t understand why the answer to this example, 195 g, is rounded to 3 significant digits instead of being rounded to 2 significant digits as 200 g. My reasoning is that multiplication is the only operation, and one of the multiplands, 70% (or alternately 70 mL / 10 mL) is only given to two significant digits. Compare with the Check Your Learning problem at the bottom of the example, where the concentration of ethanol is given as 12% and the answer, 1.5 mol, is rounded to 2 significant digits. Therefore, shouldn’t the percentage of rubbing alcohol be given as 70.0% if the answer is to be rounded to 3 significant digits, or shouldn’t the answer be rounded to 2 significant digits ?

Example 3.25, Check Your Learning problem at the bottom.
Is the 0.48 milligrams mercury given in the problem statement supposed to be 0.48 **micro**grams? With the .48 milligrams as currently given, the fraction of Mercury is 0.0096, so the answer should be 9600 ppm or 9600000 ppb, instead of the answer given, which is “9.6 ppm, 9600 ppb”.

Ch 3 Key Terms

Would it make sense to add **solution** as a key term for Chapter 3?

Ch 3 Exercises

See also the discrepancies table far above.

Ex 3.4, 3.5, and 3.6
The structural formula diagrams in these exercises are small, significantly smaller than those in the text. This is the same issue as was reported on Ch 2 exercises like 2.33(a) in a previous set of errata I submitted.

Ex 3.4(b) molecular structure image alt text
The last sentence ends with "...with **on** H atom." It should read "...with **an** H atom."

Ex 3.13(a) solution
Why are the units on the answer for 13(a) written as g / mol, whereas the answers to 13(b) through 13(e) are written as g mol^{-1}. Neither is inaccurate. I am just pointing out the inconsistency.

Ex 3.59 solution
It says .5000 L. Shouldn't it be .4999 L. The problem reads:

> If 0.1718 L of a 0.3556-M C_3H_7OH solution is diluted to a concentration of 0.1222 M, what is the volume of the resulting solution?

The dilution equation from Ch 3.3 is

$$C_1V_1 = C_2V_2$$

Plugging in the problem values gives

$$0.3556 \text{ M} \times 0.1718 \text{ L} = 0.1222 \text{ M} \times V_2$$

Solving for V_2 gives

$$V_2 = 0.3556 \text{ M} / 0.1222 \text{ M} \times 0.1718 \text{ L}$$

I have tried this on multiple calculators, and always get

> 0.499935…

Which, rounded to 4 significant digits, is

> 0.4999 L

Which is not the solution given for Ex 3.59 in the answer key, which is

> 0.5000 L

Ch 4.1

Ch 4.1, Figure 4.3 alt text (issue 1)
Three places in the alt text, sentences 1, 5, and 10, refer to the separation between the left and right sides of the diagram as a dashed (vertical) line. In fact the separation is a right-facing arrow. The three places use slightly different words each time. The three phrases to look for are "a vertical dashed line", "the dashed vertical line", and "the dashed line". For comparison, Ch 4.4 Figure 4.14 alt text involves a similar situation where the arrow is described in separate sentences or phrases reading "There is a reaction arrow pointing to the right in the middle", and "To the right of the reaction arrow".

Ch 4.1, Figure 4.3 alt text (issue 2)
There is one more inconsistency in this alt text, although it is not actually in error. Sentences 3, 4, and 6 all begin with parallel constructions, “Six of the molecules have”, “Three of the molecules have”, and “Six of the molecules have”. Sentence 8 starts the description of the final molecule type with “The second molecule type has”. Would it be clearer if sentence 8 started with a parallel construction to sentences 3, 4, and 6, such as “Three of the molecules have”?

Ch 4.2

First paragraph after Figure 4.4, 2nd-to-last sentence.
It reads “The solubility guidelines indicate all nitrate salts are soluble but that AgCl is **one of** insoluble.” The words “one of” toward the end of the sentence can be removed.

Ch 2.6, Figure 2.29, showing ionic charges, suggested additions.
Ch 4.2, Example 4.3 uses lead ions Pb2+. Since Pb is in group 14 in the periodic table, it seems non obvious that its ionic charge would be 2+ instead of 4- like C in the same group. Would it make sense to add Pb to the ionic charges table in Figure 2.29? I believe Ti was used earlier as well, so that might be another element to consider adding to Figure 2.29.

Figure 4.5 alt text (issue 1)
Sentence 5 starts “The label H subscript 2 O followed by **a q** in parentheses”. However, the parentheses contain “l” (lowercase L), not “aq”, so “a q” should be replaced with “l”.

Figure 4.5 alt text (issue 2)
Sentence 3 starting “Flask a is labeled…” and sentences 8 and 9 starting with “Flask b is labeled…” are incomplete. Sentence 3 only describes the left side of the chemical equation below flask a. Sentences 8 and 9 only describe the right side of the chemical equation below flask b. Missing are the chemical equation arrows, the right side of the flask a equation, and the left side of the flask b equation.

Figure 4.6 alt text.
Much of this alt text no longer matches the figure and should maybe be rewritten. Mismatches include:

- The alt text says the bottles are clear but they are not.
- The alt text says the label says “vinegar” but the labels are actually not in English.
- The alt text says the molecular models are in the lower left corners when they are in the upper left corners.
- The alt text goes to great lengths to describe structural formulas, including a pink highlighting, which seem to have been replaced in the image with space-filling models with no highlighting. The alt text describes much detail about the structural formulas that is not visible in the space-filling models (providing a possible opportunity to shorten the alt text.)
- Possibly more, I stopped reading after I found that much.

Figure 4.7 alt text.
The alt text describes figure 4.7(a) but omits figure 4.7(b).

Ch 4.2, Example 4.6, Check Your Learning section.
It says that the tin(II) chloride reaction is a single-replacement reaction. This is confusing because the body text above says (emphasis mine) that “Single-displacement (replacement) reactions are redox reactions in which an ion **in solution** is displaced (or replaced) via the oxidation of a metallic element.” Both example reactions in the body text involve an (aq) solution. However, the tin(II) chloride reaction in Exercise 4.6 Check Your Learning only involves a solid (s) and HCl gas (g). I don’t understand how either of those are a solution, so wanted to point this out to be checked.

Ch 4.3

Example 4.8 Solution unlabeled flowchart figure, alt text.
The alt text does not mention the “Stoichiometric factor” text appearing below the arrow.

Example 4.9 Solution unlabeled flowchart figure, alt text.
The alt text does not mention the second arrow, the third box, and the text labels below both arrows.

Example 4.10 and 4.11 Solutions, unlabeled flowchart figures, alt text.
The alt text of both of these flowcharts must have been written for previous versions. The arrow directions are wrong, and the text labels on the arrows are not mentioned.

Figure 4.11 alt text (issue 1)
Sentence 5 describing the second arrow starts `This rectangle is followed by a double headed arrow which is labeled, “Molar Mass,” that connects to a third rectangle`. Unlike the other arrow descriptions, this one does not indicate where the arrow is in relation to the previous rectangle. Adding an arrow location (below and to the right) would be more consistent. It also might avoid confusion for a reader who reads that the first arrow was horizontal, assumes the second arrow just described as “following” would also be horizontal, but then reads that the third box has a horizontal arrow to its left…right where the reader thought the second arrow was.

Figure 4.11 alt text (issue 2)
One thing that is obvious to someone looking at the figure is that the substance B right side is basically a mirror of the substance A side, divided by the arrow labelled “Stoichiometric factor.” To provide alt text readers with this shortcut to understanding, I suggest inserting a new sentence explicitly calling that out before going on to describe side B.
More specifically, I suggest modifying sentences 10 and 11. They currently read ‘To the right of the pink “Moles of A,” rectangle is a horizontal double headed arrow which is labeled, “Stoichiometric factor.” **It connects** to a second pink rectangle which is labeled, “Moles of B.”’ I

suggest that portion would now read ‘To the right of the pink “Moles of A,” rectangle is a horizontal double headed arrow which is labeled, “Stoichiometric factor.” **It connects to a right half of the diagram for substance B which mirrors the half of the diagram previously described for substance A. Specifically, the arrow labeled “Stoichiometric factor” connects** to a second pink rectangle which is labeled, “Moles of B.”’

Figure 4.11 alt text (issue 3)
Sentence 13 is missing a word in some box text. It reads ‘A horizontal double headed arrow which is labeled, “Density” links to a lavender rectangle labeled, “Volume of substance B,” to the right.’ It should read ‘A horizontal double headed arrow which is labeled, “Density” links to a lavender rectangle labeled, “Volume of **pure** substance B,” to the right.’

Figure 4.11 alt text (issue 4)
Sentence 14 reads ‘ A horizontal double headed arrow labeled, “Molarity,” extends right to the of the pink “Moles of B” rectangle.’ If the sentence is left as-is, the ‘extends right **to the of** the pink’ should read ‘extends right **from** the pink’. However, the sentence might be clearer if the clauses were reversed so that the reader has the previously mentioned “Moles of B” rectangle back in mind before describing the arrow that proceeds from it. Basically I am suggesting a construction more like sentences 7 and 8, such as ‘The pink, “Moles of B,” rectangle is also connected with a double headed horizontal arrow to the right labelled “Molarity.”’

Figure 4.11 alt text (issue 5)
Sentence 15 has a mistaken word in some box text. It reads ‘This arrow connects to a lavender rectangle that is labeled, “Volume of **substance** B.”’ It should read ‘This arrow connects to a lavender rectangle that is labeled, “Volume of **solution** B.”’

Ch 4.4

Figure 4.13 alt text.
Sentence 6 describing row 3 reads (emphasis mine) ‘The third row reads, “We can make: 11 sandwiches + 6 slices **of** bread left over.”’ The word “of” does not appear on the diagram. (I might be clearer to leave it unchanged, but it is a difference.)

Figure 4.14 alt text (issue 1)
Sentence 1 reads (emphasis mine) ‘The figure shows **a** space-filling molecular models reacting.’ The word “a” should be removed.

Figure 4.14 alt text (issue 2)
Sentence 3 reads (emphasis mine) ‘To the left of the reaction arrow there are **three** molecules each consisting of two green spheres bonded together.’ The number “three” should be “four”.

Figure 4.14 alt text (issue 3)

Sentence 4 reads (emphasis mine) ‘There are also **five** molecules each consisting of two smaller, white spheres bonded together.’ The number “five” should be “six”.

Ch 4.5

Figure 4.16 alt text (issue 1)
The alt text appears out of date. For example, part (b) describes labels of 10, 15, and 20, but the labels on the pictured glass tube are 0, 1, 2, 3.

Figure 4.16 alt text (issue 2)
Since this figure is linked earlier from the term “buret”, and since what a buret is or looks like is not explained in the test, it might also be a good idea to describe the long glass tube and tap/valve in addition to saying “a buret”. Alternatively, it might be a good idea to say what a buret is up near the bold term “buret” in the body text.

Example 4.14 Solution unlabeled flowchart.
The directions for the second and third arrows are wrong.

Figure 4.18 alt text.
Sentence 5 begins “**An** line…” It should be “**A** line…”

Example 4.16.
Both the main problem and the Check Your Learning problem give extra information not needed for the problem, specifically the 0.00126 g and 0.00215 g sizes of the samples. I haven’t seen that done elsewhere so I’m mentioning it in case it is not intentional.

Example 4.16 Solution unlabeled flowchart figure.
The fourth column of boxes in the flowchart are not relevant to the Example. They appear to be a correct extension of it, but not necessary for the problem as given.

Example 4.16 Solution unlabeled flowchart figure alt text (issue 1)
The alt text starts ‘This figure shows two flowcharts.’ However, it is not two separate flowcharts. It is one flowchart, as can be seen by the first box in the third row saying “C to H mole ratio”, where the C comes from the first row and H comes from the second row, meaning the first row is not a separate flowchart.

Example 4.16 Solution unlabeled flowchart figure alt text (issue 2)
Sentence 10 says that the third rectangle in row 1 ‘...is followed by an arrow **labeled “Molar mass”** which points right to a fourth rectangle.’ In fact, that arrow is not labeled on the flowchart.

Example 4.16 Solution unlabeled flowchart figure alt text (issue 3)

The same thing as issue 2 happens in sentence 19, which says that the third rectangle in row 2 '...is followed to the right by an arrow **labeled, "Molar mass,"** which points to a fourth rectangle.' In fact, that arrow is not labeled on the flowchart.

Example 4.16 Solution unlabeled flowchart figure alt text (issue 4)
Finally, sentence 21, starting from the fourth rectangle in row 2, says 'An arrow **labeled, "Sample mass"** points down beneath this rectangle to a green shaded rectangle.' Again, the arrow is not labeled on the flowchart.

Ch 4 Exercises

See also the discrepancies table far above.

General text wrapping.
When viewed in a narrow browser window, the text of some questions does not seem to wrap at the window width. This is especially pronounced in the text for question 25. Tested in Chrome on a landscape Android tablet with the screen split left and right. Also tested with Firefox on Windows 11 by narrowing the browser window.

Exercise 4.13 (a) answer.
The answer is given as "oxidation-reduction (addition)". I'm not sure it matters, but the term "addition" does not appear as a type of reaction in the Ch 4.2 text, nor does it appear in the Chapter 4 Key Terms.

Exercise 4.25 (b).
The problem states that the reactions are "acid-base neutralization reactions." The Ch 4.2 body text says (emphasis mine) "A neutralization reaction is a specific type of acid-base reaction in which the reactants are an acid and a base **(but not water)**," However, exercise 4.25 (b) includes water as one of the reactants. This may not affect the students ability to work the problem, and I see now that it is a reversed neutralization reaction, but am reporting it so someone more knowledgeable than me can determine if the wording is confusing.

Exercise 4.25.
The problem says (emphasis mine) "If water is used as a solvent, write the **reactants and** products as aqueous ions". However, none of the answers break down the reactants into ions, only the products. Would the 4.25 (a) reactant 2 $HClO_4$ (aq) dissolve into 2 H^+ and 2 ClO_4^+ ions in water? If so, should the words "reactants and" be removed from the problem statement?

Exercises 4.37, 4.39, and 4.41 answers (may apply to even numbered exercises as well).
As a chemistry learner, I may be missing something. However, the treatment of H^+ ions versus H_3O^+ ions in the answers for acid solutions seem inconsistent. That is, the answers seem inconsistent with each other, and some answers seem inconsistent with the half-reaction

method as described in Chapter 4.2, section *Balancing Redox Reactions via the Half-Reaction Method*.

For example, take exercise 4.39 (b). When I worked this problem I got an answer:

H_2S (g) + Hg_2^{2+} (aq) → 2 Hg (l) + S (s) + 2 H^+ (aq).

However, the answer given in the book is:

H_2S (g) + Hg_2^{2+} (aq) + 2 H_2O (l) → 2 Hg (l) + S (s) + 2 H3O^+ (aq)

The answer given matches the description of an acid given in Ch 4.2, section *Acid-Base Reactions*, paragraph 2, '...an acid is a substance that will dissolve in water to yield hydronium ions, H_3O^+.' However, the answer given does **not** seem to match the half-reaction method instructions given in Ch 4.2. Specifically, step 4 says to 'Balance hydrogen atoms by adding H^+ ions.' Step 8 contains substeps to modify the H^+ ions for a basic solution. However, there are no similar instructions to modify the H^+ ions to H_3O^+ ions in an acidic solution. H_3O^+ ions are not mentioned in the half-reaction method instructions. Nor are they mentioned in Ch 4.2 Example 4.7, *Balancing Redox Reactions in Acidic Solution*, whose answer contains H^+ ions. The exercise answers also seem inconsistent among themselves.

Here are exercises whose answers include H_3O^+ ions:

- Exercise 4.39 (b)
- Exercise 4.39 (c)
- Exercise 4.39 (e)

Here are exercises whose answers include H^+ ions:

- Exercise 4.37 (d)
- Exercise 4.37 (f)
- Exercise 4.37 (g)
- Exercise 4.41 (c)

Exercise 4.43 solutions in general, but especially 4.43 (e).

The answers are multi-part. However, different parts are included in different answers. 4.43 (e) as is the most inconsistent. I will call the side of the reaction for which the initial mass is given the "same side", and the other side of the reaction the "other side". Using that wording, the answers include:

- 4.43 (a) Same side moles, other side moles, other side mass.
- 4.43 (b) Same side moles, other side moles, other side mass.
- 4.43 (c) Other side moles, other side mass.
- 4.43 (d) Other side moles, other side mass.
- 4.43 (e) Same side moles, other side mass. (This is the odd one. The only moles and only mass are labelled with different chemical species, CuO vs $CuCO_3$.)
- 4.43 (f) Other side moles, other side mass.

Compare with Exercise 4.45, which is a similar problem, but with self-consistent answers that always give only other side moles and other side mass for the solution.

Exercise 4.61 solution.

For some reason the 2 in Cl_2 is not subscripted.

Exercise 4.69 solution.

The steps portion of the solution starts by saying ‘Convert mass of ethanol to moles of ethanol;’ However, mass of ethanol is a piece of information not provided by the problem statement. Shouldn’t there be a step even before that to to convert the volume of ethanol to mass of ethanol?

Exercise 4.77.
The problem statement directs the student to http://money.cnn.com/data/commodities/. It is not formatted as a link, but it doesn’t matter because the link is broken. Pasting that URL into a browser just shows a “Page Not Found!” error.

Exercise 4.77 solution.
The solution states (emphasis mine) ‘This amount cannot be **weighted** by ordinary balances and is worthless. ’. The word “weighted” should be “weighed” with no “t”.

Exercise 4.92 image alt text.
Sentence 8 begins ‘To the right of this arrow is a structural formula that begins C a,...’ Instead of “begins Ca” it should read “begins **2** C a”.

Thank you.

Dean Peterson - March 11, 2026